Curriculum Vitae

Personal Information



Full Name	Nessrine ELLOUMI
Date and place of birth	22/02/1991 in Sfax, Tunisia
Degree	PhD in Computer Science
Position	**Assistant Professor** at the Higher Institute of Applied Sciences and Technology of Sousse (ISSAT).
Address	85 Haffouz Street, 3000 Sfax, Tunisia
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Languages	French (B2) English (B2)
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Portfolio	www.nessrine-elloumi.tn
ORCID	0000-0002-6865-0769

Education

May 2021
National PhD Degree in Computer Science
Higher Institute of Computer Science and Communication Technologies of Hammam Sousse (ISITCom).
Title: Study and development of objective metrics to measure the perceptual quality of static 3D models.
Distinction: **Very Honourable (highest distinction)**

June 2015
National Engineering Degree in Applied Sciences and Technology
Major: Computer Science, Web and Multimedia Technology
Higher Institute of Computer Science and Multimedia of Sfax (ISIMS)
Distinction: **Pass**

July 2014
National Professional Master's Degree
Major: Computer Networks and Telecommunications
National School of Electronics and Telecommunications of Sfax (Enet'Com)
Distinction: **Fairly Good**

June 2012
National Applied Bachelor's Degree in Computer Technology
Major: Networks and IT Services
Higher Institute of Technological Studies of Sfax (ISET)
Distinction: **Very Good**

June 2009
National Baccalaureate in Economics and Management
Majida BOULILA Secondary School, Sfax
Distinction: **Pass**

Scientific Publications

Journal Articles

1 Nashmi Hassan ALRASHEEDI, Mehdi TLIJA, **Nessrine ELLOUMI** and Borhen LOUHICHI, "A critical review of 3D printed orthoses towards workflow implementation in the clinical practice", Journal of Engineering Research **(2025)**
(Scopus, Q3, IF: 1)

DOI: 10.1016/j.jer.2024.01.024

2 Sabrine BEN AMOR, **Nessrine ELLOUMI,** Ameni ELTAIEF, Borhen LOUHICHI, Nashmi H. Alrasheedi and Abdennour Seibi, "Digital Twin Implementation in Additive Manufacturing: A Comprehensive Review", Processes, **(2024)**
(Scopus, Q2, IF: 3.4)

DOI: 10.3390/pr12061062

3 **Nessrine ELLOUMI,** Habiba LOUKIL, and Med Salim BOUHLEL, "Full-reference objective quality metric for three-dimensional deformed models", International Journal of Image and Graphics, Vol. 24, No. 01, 2450006 **(2024)**
(Scopus, Q3, IF: 1.6),

DOI: 10.1142/S0219467824500062

4 **Nessrine ELLOUMI**, Aicha BEN MAKHLOUF, Nuha Mohammed Alshuqayran and Borhen LOUHICHI, "Enhancement of B-Spline surface rebuilding for deformed CAD model reconstruction", International Journal of Modeling, Simulation, and Scientific Computing **(2024)**
(Scopus, Q4, IF: 0.8)

DOI: 10.1142/S1793962325500059

5 Aicha BEN MAKHLOUF, **Nessrine ELLOUMI**, Borhen LOUHICHI and Mehdi JAIDANE, Nashmi Hassan ALRASHEEDI, and Kholoud SANDOUGAH, "Biomechanical modeling and simulation of a human organ using an Augmented Reality technique during open surgery", International Journal of Modeling, Simulation, and Scientific Computing, Vol. 14, No. 05, 2350045 **(2023)**
(Scopus, Q4, IF: 1.2)
DOI: 10.1142/S1793962323500459

6 **Nessrine ELLOUMI**, Aicha BEN MAKHLOUF, Ayman AFLI, Borhen LOUHICHI, Mehdi JAIDANE and João Manuel R. S. TAVARES, "CT images segmentation using Deep learning approach for preoperative projection of human organ model using Augmented Reality technology", International Journal of Computational Intelligence and Applications, Vol. 22, No. 02, 2350006 **(2023)**
(Scopus, Q3, IF: 1.8)
DOI: 10.1142/S1469026823500062

7 **Nessrine ELLOUMI**, Habiba LOUKIL, Nilanjan DEY, Amira S. ASHOUR, and Med Salim BOUHLEL, "Perceptual Metrics Quality: Comparative Study for 3D Static Meshes", International Journal of Service Science, Management, Engineering, and Technology (IJSSMET), Vol.8, no.1 p 63-80 (**2017)**
(Scopus)
DOI: 10.4018/IJSSMET.2017010105

International Conference Papers

1 Ayman AFLI, **Nessrine ELLOUMI**, Aicha BEN MAKHLOUF, Borhen LOUHICHI, Mehdi JAIDANE, and João Manuel RS TAVARES. "Preoperative Image Segmentation for Organ Visualization Using Augmented Reality Technology During Open Liver Surgery." In 2022 26th International Conference Information Visualisation (IV 2022), Vienna, Austria, 19-22 July 2022, pp. 442-445. **(Class B)**
Published: 23 January **2023**
DOI: 10.1109/IV56949.2022.00078

2 Aicha BEN MAKHLOUF, Anass AYED, **Nessrine ELLOUMI**, Borhen LOUHICHI, Mehdi JAIDANE, and João Manuel Rs TAVARES. "Biomechanical Modeling and Pre-Operative Projection of A Human Organ using an Augmented Reality Technique During Open Hepatic Surgery" In 2022 26th International Conference Information Visualisation (IV 2022), Vienna, Austria, 19-22 July 2022, pp. 446-450. **(Class B)**
Published: 23 January **2023**
DOI: 10.1109/IV56949.2022.00079

3 **Nessrine ELLOUMI,** Habiba LOUKIL, and Med Salim BOUHLEL, "Application of the Laplacian Smoothing on 3D Static Models and Their Evaluation by the New Objective Quality Metric 3DrwPSNR", 14th International Conference on Computational Collective Intelligence (ICCCI 2022), In Advances in Computational Collective Intelligence. Communications in Computer and Information Science, Hammamet, Tunisia, 28-30 September 2022, Vol 1653. pp. 305-314. 2022, **(Class B)**
Published: 21 September **2022**
DOI: 10.1007/978-3-031-16210-7_25

4 **Nessrine ELLOUMI**, Aicha Ben Makhlouf, Borhen Louhichi and Dominique Deneux, "Towards a building technique of a BREP model starting from a meshed surface", International Design Engineering Technical Conferences & Computers and Information in Engineering Conference (IDETC-CIE 2021), Online, 17-20 August 2021, Vol. 85376, p V002T02A006 **(Scopus)**
Published: 17 November **2021**
DOI: 10.1115/DETC2021-71724

5 Aicha Ben Makhlouf, Nessrine ELLOUMI, Borhen Louhichi and Dominique Deneux, "Approach for CAD model reconstruction basing on 3D points insertion and surface approximation", 25th International Conference Information Visualisation (ICIV'21), Online, 5-9 July 2021, p 310-319. (Class B)

Published: 28 October 2021
DOI: 10.1109/IV53921.2021.00057

6 **Nessrine ELLOUMI**, Johan Debayle, Habiba LOUKIL, and Med Salim BOUHLEL, "A locally weighted metric for measuring the perceptual quality of 3D objects", 20th Intelligent Systems Design and Applications (ISDA'20), Online, December 12-15, 2020, p 128-139. **(Class C)**
Published: 03 June **2021**
DOI: 10.1007/978-3-030-71187-0_12

7 **Nessrine ELLOUMI,** Habiba LOUKIL, and Med Salim BOUHLEL, "A comparative study of the 3D quality metrics: application to masking Database", 18th Intelligent Systems Design and Applications (ISDA'18), Vellore, India, December 6-8, 2018 p. 740-748. **(Class C)**
Published: 14 April **2019**
DOI: 10.1007/978-3-030-16660-1_72

8 **Nessrine ELLOUMI**, Habiba LOUKIL, and Med Salim BOUHLEL, "Quality Metric of 3D Models Using Weber's Law", 11th International Conference on Machine Vision (ICMV'18), Munich, Germany, November 1-3, 2018, p. 110410B. **(Class C)**
Published: 15 March **2019**
DOI: 10.1117/12.2522846

9 **Nessrine ELLOUMI,** Habiba LOUKIL, and Med Salim BOUHLEL, "The efficiency of perceptual quality metrics 3D meshes based on the human visual system", 7th International Workshop on Soft Computing Applications (SOFA'16), Arad, Romania, August 24–26, 2016, p. 497-508. **(Class C)**
Published: 02 September **2017**
DOI: 10.1007/978-3-319-62521-8_43

Scientific Outreach

Scientific Committees

1 **Participation** in the reviewing committee of the journal *« Pattern Analysis and Applications Journal »*, Springer ***IF : 2.580, Q2***

2 **Participation** in the reviewing committee of the journal *« International Journal of Computational Intelligence and Applications »*, World Scientific Publishing**, *IF : 1.8, Q3***

3 **Participation** in the scientific committee of the international conference « *IEEE SETIT 2022* », ***IEEE***

Research Projects (Ongoing)

- **Augmented Reality and Additive Manufacturing in medical surgery:** In collaboration with Sahloul Hospital and the Higher Institute of Applied Sciences and Technology of Sousse.
  Website: www.am-g.tn
- **From 3D Scan to Printed Medical Orthotic device 3DSPMed:** In collaboration with Imam Mohammad Ibn Saud Islamic University (IMSIU), Saudi Arabia.

Research Internship

Laboratory: SPIN Center, École des Mines de Saint-Étienne, Saint-Étienne, France

Supervised by: Professor Johan DEBAYLE

Topic: Development of an objective perceptual metric for static 3D meshes

Duration: *Two months* ***(01/05/2019 – 28/06/2019)***

Funding: Alternating-mobility scholarship

Scientific Distinctions

Workshops and Sessions Led

1 **Session co-chair:** Signal and image processing (SIP)
International Conference: *«* 2022 IEEE 21st International conference on Sciences and Techniques of Automatic control & computer engineering (STA'2022) »
Date: 20 December 2022, Sousse, Tunisia.

2 ***Trainer*** at the International Institute of Technology (IIT)
Workshop: *Smart Network Simulation, from Concept to Practice (IoT)*
Date: 03 December 2021, Sfax, Tunisia

3 ***Trainer*** at the Career and Skills Certification Center of the Higher Institute of Applied Sciences and Technology of Sousse (ISSAT Sousse)
Workshop: *MATLAB training "Matlab Day".*
Date: 21 June 2021, Sousse, Tunisia

4 ***Trainer*** at the International Institute of Technology (IIT)
Workshop: *Smart Network Simulation, from Concept to Practice (IoT).*
Date: 28 February 2020, Sfax, Tunisia

Organising Committees

1 ***Organising Committee Member*** of the international conference *«9th SETIT 2022 »*, 28-30 May 2022, Sfax, Tunisia.

2 ***Member of the accreditation file preparation committee*** for the engineering programme at the Higher Institute of Applied Sciences and Technology of Sousse (ISSAT), 2022.

3 ***Organising Committee Member of the event*** « *24 Hours of Innovation* », ISSAT Sousse, 10-11 December 2021.

4 ***Organising Committee Member of the Technical Day*** on Biomechanics « *JTBM 2021* », 18 July 2021 (online).

5 ***Organising Committee Member of the Technical Day*** on *« Metal Additive Manufacturing »*, Technological Resources Center CRT-Sousse, 24 June 2021.

6 ***Organising Committee Member*** of the international workshop *« Machine Learning & Python »*, 20-22 December 2019, Monastir, Tunisia.

7 ***Organising Committee Member*** of the international conference *«8th SETIT 2018 »*, 18-20

	December 2018, Hammamet, Tunisia.
8	***Organising Committee Member*** of the 26th SETIT Unit Days *« Deep Learning Training »*, 28-30 March 2018, Sousse, Tunisia.
9	***Organising Committee Member*** of the international conference *«7th SETIT 2016 »*, 18-20 December 2016, Hammamet, Tunisia.
10	***Organising Committee Member*** of the 27th SETIT Unit Days *« LaTeX Training »*, 20-22 December 2015, Sousse, Tunisia.

Participation in Scientific Events

1	***Participation*** in the international conference *«2022 IEEE 21st International conference on Sciences and Techniques of Automatic control & computer engineering (STA'2022) »*, 19-21 December 2022, Sousse, Tunisia.
2	***Participation*** in the training *« LATEX »*, Faculty of Sciences of Sfax, 20-24 February 2019.
3	***Participation*** in the workshop *« Manage your bibliographic references with Zotero »*, ISB, Sfax, 20 February 2019.
4	***Participation*** in the training *« English B1+ »*, Bourguiba Institute of Modern Languages of Sfax, July 2018.
5	***Participation*** in the training *« Deep Learning : Theory and applications »*, 01 December 2018, Sousse, Tunisia.
6	***Participation*** in the training « *MATLAB* », Faculty of Sciences of Sfax, 20-24 November 2018.
7	***Participation*** in the workshop « *Introduction to the H2020 Research and Innovation Programme* », 18 April 2017, Sousse, Tunisia.
8	***Participation*** in the « *1st edition of the Tunisian Computer Science Research Day* », 18 April 2017, Sousse, Tunisia.
9	***Participation*** in the « *1st edition of the Young Researchers Day JJC'17* », 15 April 2017, Sousse, Tunisia.
10	***Participation*** in the training « *Web of Science and EndNote* », Faculty of Medicine of Sfax, November 2017.
11	***Participation*** in the training « *Android Development* » at the Higher Institute of Computer Science and Multimedia of Sfax, from 29 June 2015 to 03 July 2015.

Teaching Activities

Teaching Experience

2025-Present **<u>Assistant Professor</u>** at the Higher Institute of Applied Sciences and Technology of Sousse (ISSAT).

- Lectures, Labs: Artificial Intelligence
  3rd year, Bachelor in Computer Systems Engineering
- Lectures, Tutorials: IP Networks
  1st year, Research Master in Computer Science: Intelligent Pervasive Systems (SPI)

- Lectures: Portal Development and Collaborative Work Tools
  2nd year, Computer Engineering programme

2023-2024

Contract Assistant Professor (PhD Contract) at the Higher Institute of Industrial Management of Sfax (ISGI).

- Lectures, Tutorials, Labs: Mobile Programming
  3rd year, Bachelor in Logistics Engineering
- Lectures: Mobile Development
  3rd year, Bachelor in Business Computing
- Labs: Project Management
  3rd year, Bachelor in Business Computing
- Labs: Industrial Local Area Networks
  3rd year, Bachelor in Electrical Engineering and Automation
- Integrated Course: C2i
  1st year, Bachelor in Transport Sciences
- Labs: Web Programming 2.
  2nd year, Bachelor in Business Computing

Adjunct Lecturer at the National School of Electronics and Telecommunications of Sfax (Enet'Com).

- Labs: Artificial Intelligence and Machine Learning
  2nd year, Professional Master in Electronics, Electrical Engineering and Automation.
- Labs: Image Processing
  2nd year, Industrial Computer Engineering programme.

2022-2023

Contract Assistant Professor (PhD Contract) at the Higher Institute of Applied Sciences and Technology of Sousse (ISSAT).

- Lectures, Tutorials: Software Design Methodologies
  2nd year, Bachelor in Computer Systems Engineering
- Lectures, Tutorials: Software Testing, ISTQB Certification Preparation
  2nd year, Bachelor in Computer Systems Engineering
- Lectures: Portal Development and Collaborative Work Tools
  2nd year, Computer Engineering programme

Adjunct Lecturer at the National School of Electronics and Telecommunications of Sfax (Enet'Com).

- Labs: Artificial Intelligence and Machine Learning
  2nd year, Professional Master in Electronics, Electrical Engineering and Automation.
- Labs: Image Processing.
  2nd year, Industrial Computer Engineering programme

2021-2022

Contract Assistant Professor (PhD Contract) at the Higher Institute of Applied Sciences and Technology of Sousse (ISSAT).

- Lectures: Algorithms and Data Structures 1
  1st year, Bachelor in Computer Systems Engineering
- Lectures, Labs: Portal Development and Collaborative Work Tools
  2nd year, Computer Engineering programme

- Tutorials: Computer Networks
  2nd year, Integrated Scientific Preparatory Cycle: Computer Science

2020-2021

Adjunct Lecturer at the Higher Institute of Applied Sciences and Technology of Sousse (ISSAT).

- Lectures: Algorithms and Data Structures 1
  1st year, Integrated Scientific Preparatory Cycle: Computer Science
- Lectures: Portal Development and Collaborative Work Tools
  2nd year, Computer Engineering programme

Adjunct Lecturer at the Higher Institute of Industrial Management of Sfax (ISGI).

- Labs: Web Programming 2.
  2nd year, Bachelor in Business Computing: Business Intelligence.

Adjunct Lecturer at the National School of Electronics and Telecommunications of Sfax (Enet'Com).

- Labs: Image Processing.
  2nd year, Industrial Computer Engineering programme.

2019-2020

Contract Assistant Professor (PhD Student Contract) at the National School of Electronics and Telecommunications of Sfax (Enet'Com).

- Labs: Data Transmission and Protocols Workshop
  3rd year, Fundamental Bachelor in Electronics
  3rd year, Fundamental Bachelor in Telecommunications
- Labs: Switching and Routing CCNA 2
  2nd year, Telecommunications Engineering programme
- Labs: Quality of Service in IP Networks
  2nd year, Professional Master in Computer Networks and Telecommunications
- Labs: Telecom Architecture and Networks
  1st year, Electronic and Communication Systems Engineering programme
- Labs: Networking Workshop
  1st year, Telecommunications Engineering programme
- Labs: Network Certification Preparation
  1st year, Professional Master in Computer Networks and Telecommunications

2018-2019

Adjunct Lecturer at the Higher Institute of Industrial Management of Sfax (ISGI).

- Labs: Networking Workshop
  2nd year, Applied Bachelor in Electrical Engineering and Automation

Adjunct Lecturer at the Higher Institute of Biotechnology of Sfax (ISBS).

- Labs: Digital Processing of Medical Signals.
  3rd year, Applied Bachelor in Biomedical Engineering

Teaching Materials

2022-2023

Lab Manual: Image Processing.
2nd year, Industrial Computer Engineering programme.

2021-2022	***Lab Manual:*** Portal Development and Collaborative Work Tools *2nd year, Computer Engineering programme.*
2021-2022	***Lecture Notes:*** Algorithms and Data Structures 1 *1st year, Integrated Scientific Preparatory Cycle*
2020-2021	***Lab Manual:*** Web Programming 2 *2nd year, Bachelor in Business Computing: Business Intelligence*
2019-2020	***Lab Manual:*** Quality of Service in IP Networks *2nd year, Professional Master in Computer Networks and Telecommunications*
2018-2019	***Lab Manual:*** Networking Workshop *2nd year, Applied Bachelor in Electrical Engineering and Automation*
2018-2019	***Lab Manual:*** Digital Processing of Medical Signals *3rd year, Applied Bachelor in Biomedical Engineering*

Supervision of Final-Year Projects (Bachelor)

1	***Title:*** Business tracking and KPI computation platform *(ongoing)* at ISGI Sfax
2	***Title:*** Automatic vehicle classification *(ongoing)* at ISGI Sfax
3	***Title:*** Conformity detection of sugared-almond pieces *(ongoing)* at ISGI Sfax
4	***Title:*** Mobile application for the SONEDE customer portal *(ongoing)* at ISGI Sfax
5	***Title:*** Development of a platform for nucleus detection and counting ***Defended on:*** 03/07/2023 at Enet'Com Sfax
6	***Title:*** Design and implementation of the services platform: Informini ***Defended on****:* 23/06/2023 at ISSAT Sousse
7	***Title:*** Design and implementation of a stadium management system ***Defended on:*** 23/06/2023 at ISSAT Sousse
8	***Title:*** Setting up an employability observatory for ISSAT alumni ***Defended on:*** 30/06/2022 at ISSAT Sousse

Supervision of Final-Year Projects (Engineering)

1	***Title:*** Design and development of a web platform for human resources management « Talenteo » ***Defended on:*** 13/09/2023 at Enet'Com Sfax
2	***Evaluation of 8 final-year projects:*** Jury President and Examiner for engineering final-year projects (2023) at ISSAT Sousse.
3	***Title:*** Design and development of an augmented reality application for magnetic field

	visualisation ***Defended on:*** 28/10/2023 at ISSAT Sousse
4	***Title:*** Design and development of a mobile application for managing and tracking test cases ***Defended on:*** 14/10/2023 at ISSAT Sousse
5	***Title:*** Design and implementation of a mobile application for parcel shipping and tracking ***Defended on:*** 11/07/2023 at ISSAT Sousse
6	***Title:*** Design and development of a mobile application for a high-end products marketplace. ***Defended on:*** 07/07/2023 at ISSAT Sousse
7	***Title:*** Design and development of a delivery platform ***Defended on:*** 11/07/2023 at ISSAT Sousse

Co-supervision of Research Master's Theses

1	***Title:*** Biomechanical modelling and 3D reconstruction of an organ for augmented reality simulation during open surgery. ***Defended on:*** 27/11/2021 at ISSAT Sousse

Certifications and Software Tools

23/01/2024	Cisco Certified Network Associate **CCNA 3**: Enterprise Networking, Security and Automation.
11/01/2023	Microsoft Azure Artificial Intelligence Fundamentals
26/12/2021	Cisco Certified Network Associate **CCNA 2**: Switching, Routing and Wireless
23/01/2020	Cisco Certified Network Associate **CCNA 1**: Introduction to Networks

Development Methodologies

05/07/2021	Scrum master Certification (**SMC**)
12/06/2021	Scrum Fundamentals certification (**SFC**)